Supplementary material for *Dynamic consequences of declining vaccination coverage in a heterogeneous world*

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1 Theoretical honeymoon time in the SIR model

We consider here how reductions in vaccination can lead to rebounds in disease prevalence in a SIR (Susceptible, Infected, Recovered) model. In particular, we will focus on the post-vaccination "honeymoon", or how long it takes for the disease to rebound after vaccination starts declining. Note, we use the term "immunization" thorughout to account for both vaccination coverage and imperfect vaccine efficacy (i.e., immunization represents those successfully immunized, not merely vaccinated).

To begin, we will consider the simplest case where (a) a population has been under a "pre-decline" vaccination regime for a sufficiently long period of time to be at equilibrium, and (b) that the pathogen is locally eliminated. Let v_0 be the immunization rate during the period of herd immunity, and let v be the immunication rate after the decline begins.

In the SIR model, we know there are two possible equilibria:

1. disease-free equilibrium, where $S^* = v_0$
2. endemic equilibrium, where $S^* = \frac{1}{\beta}(\mu + \gamma + \frac{\mu(\mu+\gamma)}{\sigma})$

Here, we only consider the first case where a population is above the herd immunity threshold, $1/R_0$, because otherwise there is no honeymoon period. We are interested in the duration of the honeymoon period, which we hypothesize will depend on a number of factors including the R_0 of the pathogen, the equilibrium number of susceptibles S^*, the birth rate μ, the immunization rate v under decline, and the timing of importations. Because we cannot control the timing of importations, we define the honeymoon period as the time it takes for $R_e = R_0 \cdot S(t) > 1$. In other words, the time it takes for R_e to cross 1 is a lower bound on the honeymoon period, where any importation after this point will start an outbreak.

We can calculate analytically the time it takes for R_e to cross 1, under assumptions of local extinction (i.e., $I = 0$) and a constant post-decline immunization rate, v. In this case, we have a differential equation for the number of susceptibles, S, which we can solve using the following steps:

$$\frac{dS}{dt} = \mu(1-v) - \mu S$$
$$e^{\mu t}\Big(\frac{dS}{dt} + \mu S\Big) = e^{\mu t}\mu(1-v)$$
$$\int e^{\mu t}\Big(\frac{dS}{dt} + \mu S\Big)dt = \int e^{\mu t}\mu(1-v)dt$$
$$\int e^{\mu t}\frac{dS}{dt} + e^{\mu t}\mu S dt = \int e^{\mu t}\mu(1-v)dt$$
$$e^{\mu t}S = (1-v)e^{\mu t} + C$$
$$S(t) = (1-v) + Ce^{-\mu t}$$

and solving $S_0 = (1-v) + Ce^{-\mu \cdot 0}$ gives us $C = S_0 - (1-v)$. Thus, we have:

$$S(t) = (1-v) + (S_0 - (1-v))e^{-\mu t}$$

We can then find the time t at which $R_e = R_0 S(t) = 1$, which yields

$$t = \frac{-1}{\mu}\log\Big(\frac{1/R_0 - (1-v)}{S_0 - (1-v)}\Big)$$

We can also rewrite this equation to show us the relationship between pre- and post-decline immunization levels that yield equivalent honeymoon periods. When vaccination rates are above the herd immunity threshold, $S_0 = 1 - v_0$ is the pre-drop vaccination coverage, so

$$t = \frac{-1}{\mu}\log\Big(\frac{1/R_0 - (1-v)}{(1-v_0) - (1-v)}\Big)$$

which can be simplified to

$$v_0 = v(1 - e^{t\mu}) + e^{t\mu}(1 - 1/R_0).$$

2 Age-structured model

Our age-structured model defines a discrete set of A age classes. Transmission between individuals of different age classes is governed by a who-acquires-infection-from-whom (WAIFW) matrix, $\theta_{i,j}$, where i and j index age classes. The full system of ordinary differential equations is as follows:

$$\frac{dS_{i=1}}{dt} = \mu(1-v) - \sum_{j=1}^{A} \theta_{1,j} S_1 I_j - \mu S_1 - \alpha_1 S_1$$
$$\frac{dS_{i\neq1}}{dt} = \alpha_{i-1} S_{i-1} - \sum_{j=1}^{A} \theta_{i,j} S_i I_j - \mu S_i - \alpha_i S_i$$
$$\frac{dI_{i=1}}{dt} = \sum_{j=1}^{A} \theta_{1,j} S_1 I_j - (\gamma+\mu) I_1 - \alpha_1 I_1$$
$$\frac{dI_{i\neq1}}{dt} = \alpha_{i-1} I_{i-1} + \sum_{j=1}^{A} \theta_{i,j} S_i I_j - (\gamma+\mu) I_i - \alpha_i I_i$$
$$\frac{dR_{i=1}}{dt} = \mu v + \gamma R_1 - \mu R_1 - \alpha_1 R_1$$
$$\frac{dR_{i\neq1}}{dt} = \alpha_{i-1} R_{i-1} + \gamma R_i - \mu R_i - \alpha_i R_i$$

where α_i is the rate of aging out of age class i, μ is the birth and death rate, γ is the recovery rate, v is the immunization rate. This model assumes a total population size of 1, and compartments represent the proportion of individuals in each age class.

We can calculate the basic reproduction number, R_0, and the effective reproduction number, R_e, for this model using the Next Generation Matrix method. In this case, the next generation matrix, K, is given by

$$K = \frac{\beta}{\gamma+\mu} \begin{bmatrix} \theta_{1,1}S_1(t) & \theta_{1,2}S_1(t) & \cdots & \theta_{1,n}S_1(t) \\ \theta_{2,1}S_2(t) & \theta_{2,2}S_2(t) & \cdots & \theta_{2,n}S_2(t) \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{n,1}S_n(t) & \theta_{n,2}S_n(t) & \cdots & \theta_{n,n}S_n(t) \end{bmatrix} = \frac{\beta}{N(\gamma+\mu)} \theta \overrightarrow{S(t)},$$

for which we calculate the dominant eigenvalues numerically.

3 Unity beta calculation

To quantify the role of age-structured dynamics, we will use a theoretical quantity called "unity beta". Consider an age-structured transmission process that is governed by heterogeneous contacts between age groups, where the number of new infections in the age structured model is

$$\text{age-structured infections} = \frac{1}{N} \sum_{i=1}^{n} \sum_{i=j}^{n} \theta_{i,j} S_i I_j$$

where $\theta_{i,j}$ is the transmission rate between age groups i and j. We can then calculate a time-varying quantity, $k(t)$, called "unity beta" that ensures the number of new infections in the age-structured model is equal to the number of new infections in a non-age-structured model, where

$$
\begin{aligned}
\text{age-structured infections} &= \frac{1}{N}\sum_{i=1}^{n}\sum_{i=j}^{n}\theta_{i,j}S_iI_j \\
&= \frac{\hat{\beta}(t)}{N}\sum_{i=1}^{n}S_i\sum_{i=j}^{n}I_j \\
&= \text{homogenous infections}
\end{aligned}
$$

given that

$$
\hat{\beta}(t) = \frac{\sum_{i=1}^{n}\sum_{i=j}^{n}\theta_{i,j}S_iI_j}{\sum_{i=1}^{n}S_i\sum_{i=j}^{n}I_j}
$$

This ratio can be interpreted as the ratio of the number of new infections in the age-structured model to the number of new infections in the homogenous model. If $k(t) > 1$ then age-structured dynamics are fast relative to the homogenous dynamics, and vice versa if $k < 1$.

4 Metapopulation model

We consider a metapopulation model with P identical patches that are assumed to be equally connected, such that $c\%$ of transmission arises from cross-patch contact. The model can be defined as

$$
\begin{aligned}
\frac{dS_p}{dt} &= \mu(1-v_p) - \beta\sum_{q=1}^{P}C_{p,q}S_pI_q - \mu S_p \\
\frac{dI_p}{dt} &= \beta\sum_{q=1}^{P}C_{p,q}S_pI_q - (\gamma+\mu)I_p \\
\frac{dR_p}{dt} &= \mu v_p + \gamma I_p - \mu R_p
\end{aligned}
$$

where $C_{p,q}$ is the connectivity matrix, which has c along the diagonal and $(1-c)/(P-1)$ elsewhere. Analagous to the age-structured case, the next generation matrix is given by

$$
K = \frac{\beta}{N(\gamma+\mu)}C\overrightarrow{S(t)},
$$

where $\overrightarrow{S(t)}$ is the number of susceptibles in each patch at time t. We specify immunization levels to drop in $\rho\%$ of these patches and the others remain at pre-decline levels.

5 Supplementary Figures

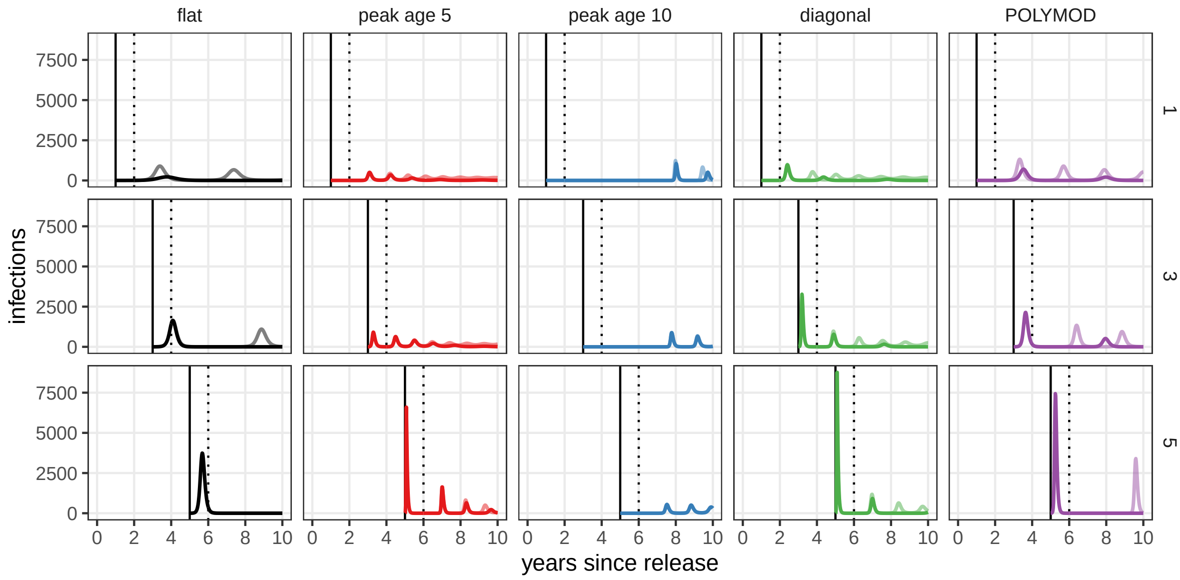

Figure S1: Sensitivity of results to the timing of infection introduction. [**ADD HERE**]

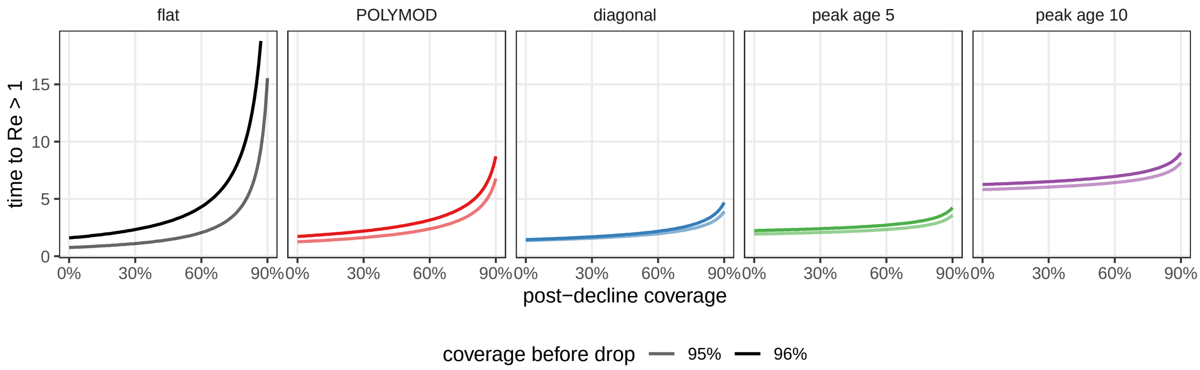

Figure S2: Theoretical honeymoon time predictions, computed as time until Re ¿ 1, under five assumptions about age-structured mixing. Results are shown for a population with $\mu = 1/80\text{years}^{-1}$, a pathogen with $R_0 = 17$, and a pre-decline immunization of 95% and 96%.

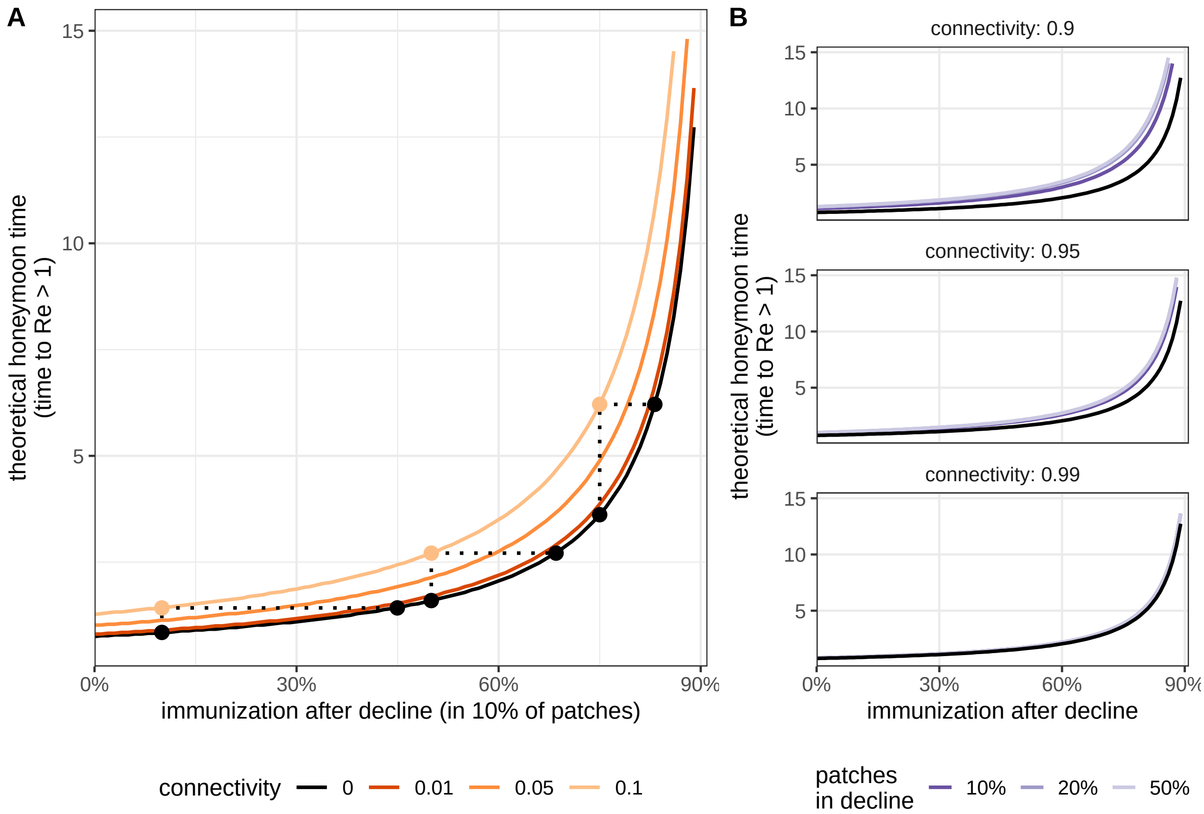

Figure S3: Role of patch connectivity in theoretical honeymoon time. (A) Theoretical honeymoon time, computed as time until Re ¿ 1, by post-decline immunization when drops occur in 10% of patches among an equally connected network. Line color shows connectivity between patches (where 0.05 indicates 5% of transmission is shared between patches, or c above), and the points show equivalence between a model with no connectivity and a model where 10% of transmission is shared between patches. (B) Sensitivity of results to the percent of patches in which immunization declines, or ρ above. Results are shown for a population with $\mu = 1/80\text{years}^{-1}$, a pathogen with $R_0 = 17$, and a pre-decline immunization of 95%.